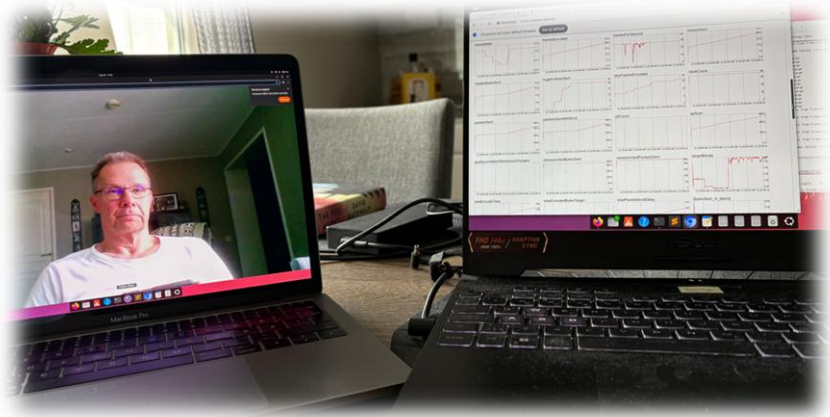


WebRTC with a local server

This smallish document outlines how to test WebRTC with a local jitsi server on a LAN and thus allow full control of selected video coder and bitrate range. The test is with the newly developed/integrated SReAM congestion



control algorithm with L4S support. The steps below are with support from Kiro so some of them may be wrong or sub-steps are missing.

Setup and run docker engine

Install docker engine

```
sudo apt update
sudo apt install ca-certificates curl gnupg
```

```
sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o
/etc/apt/keyrings/docker.gpg
sudo chmod a+r /etc/apt/keyrings/docker.gpg
```

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg]
https://download.docker.com/linux/ubuntu $(. /etc/os-release && echo
"$VERSION_CODENAME") stable" | sudo tee
/etc/apt/sources.list.d/docker.list > /dev/null
```

```
sudo apt update
sudo apt install docker-ce docker-ce-cli containerd.io docker-compose-plugin
```

Docker permissions

```
sudo groupadd docker
sudo usermod -aG docker $USER
newgrp docker
```

Clone and configure jitsi

```
git clone https://github.com/jitsi/docker-jitsi-meet
cd docker-jitsi-meet
cp env.example .env
./gen-passwords.sh
```

Create config directories and fix permissions

```
mkdir -p
~/jitsi-meet-cfg/{web,transcripts,prosody/config,prosody/prosody-plugins-
custom,prosody/storage/prosody,jicofo,jvb,ji
gasi,jibri}
sudo chown -R 1000:1000 ~/jitsi-meet-cfg
sudo chmod -R 775 ~/jitsi-meet-cfg
```

Configure .env for LAN access

Replace 192.168.x.x with your server's LAN IP (hostname -I | awk '{print \$1}')

```
sed -i 's|^#PUBLIC_URL=.*|PUBLIC_URL=https://192.168.x.x|' .env
sed -i 's|^#HTTPS_PORT=.*|HTTPS_PORT=443/' .env
sed -i 's|^#HTTP_PORT=.*|HTTP_PORT=80/' .env
sed -i 's|^#JVB_ADVERTISE_IPS=.*|JVB_ADVERTISE_IPS=192.168.x.x|' .env
```

Configure for H264 and higher bitrates, range [100kbps .. 10Mbps]

1. Edit .env file in ~/docker-jitsi-meet

Add

```
ENABLE_CODEC_H264=1
```

1. Create or edit ~/jitsi-meet-cfg/web/custom-config.js

```
config.videoQuality = {
  preferredCodec: 'H264',
  disabledCodec: 'VP8',
  disabledCodec: 'VP9',
  disabledCodec: 'AV1',
  maxBitratesVideo: {
    H264: {
      low: 200000,
      standard: 1000000,
      high: 10000000,
      ssHigh: 2500000
    }
  }
}
```

```
        }
    }
};

config.constraints = {
    video: {
        height: { ideal: 1080, max: 1080, min: 180 }
    }
};
```

Start jitsi

`docker compose up -d`

Run chrome web browsers

`chromium --vmodule=scream_v2=1 --force-fieldtrials="WebRTC-RFC8888CongestionControlFeedback/Enabled,offer:true/WebRTC-Bwe-ScreamV2/Enabled/" https://192.168.X.X/l4stest`

Results

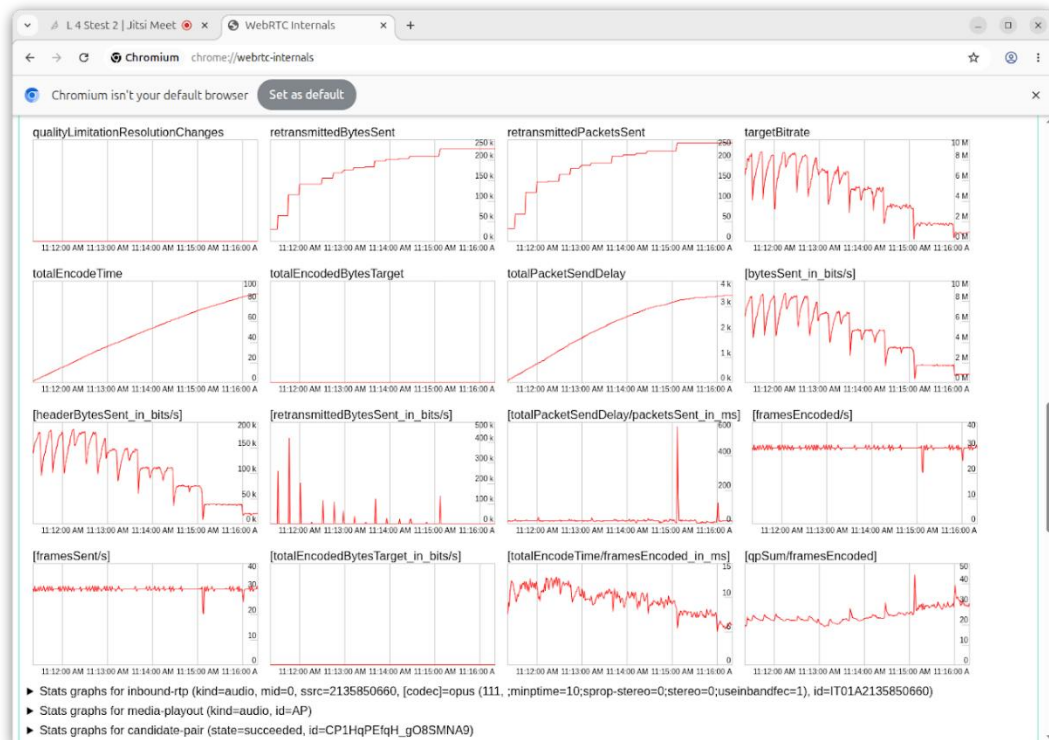
Chrome version Version 148.0.7778.167 (Official Build) snap (64-bit)

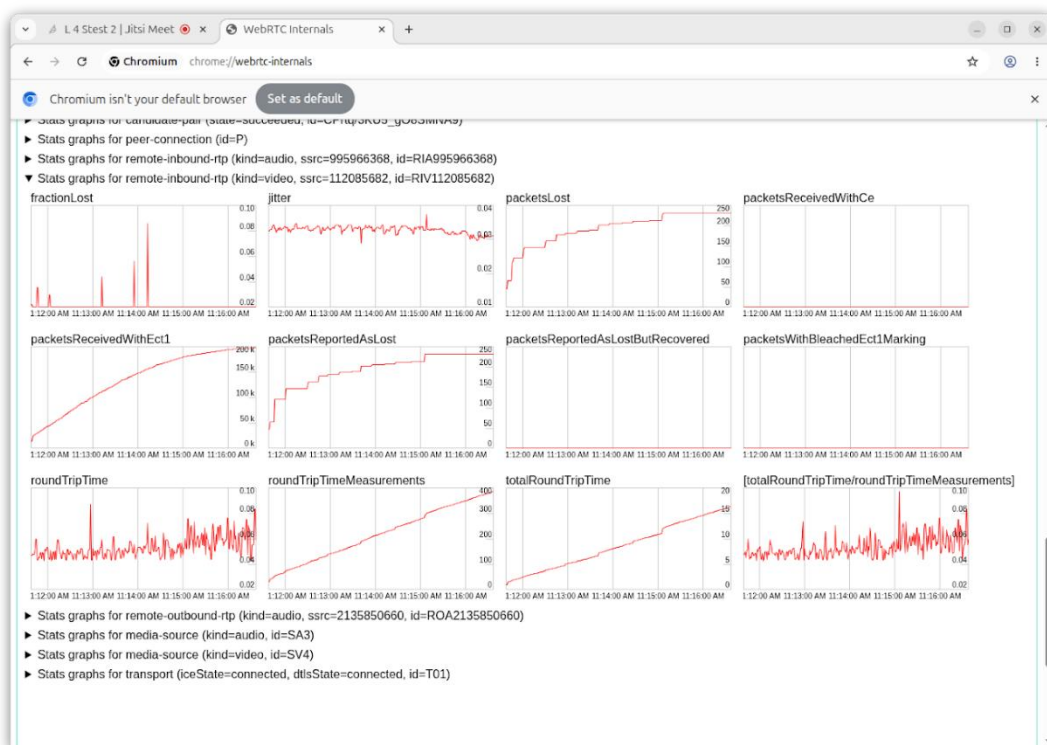
Ubuntu 24.04.4 LTS

Linux 6.17.0-23-generic

Simple link with bandwidth limitations 10-8-6-4-2-1 Mbps, 25ms RTT

L4S off





L4S on

fq_codel with 8ms ce_threshold



